

BARKHAN, PakistanWMO: **416850**Lat: **29.883N**Lon: **69.717E**Elev: **1098**StdP: **88.81**Time Zone: **5.00 (E05)**Period: **02-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.6	3.8	-15.4	1.1	13.1	-12.6	1.4	13.3	2.7	11.8	2.3	12.3	0.4	N/A	0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.7	39.0	20.0	37.7	20.2	36.5	20.2	25.2	33.2	24.6	32.6	24.1	32.0	1.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	20.5	29.9	22.6	19.8	29.5	22.0	19.1	29.0	84.2	33.4	81.7	32.3	79.4	32.0	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
3.4	2.5	2.0	1.1	41.2	1.7	1.2	-0.1	42.1	-1.1	42.8	-2.1	43.5	-3.4	44.4	
			DB												
			WB	-2.1	26.6	1.1	0.9	-2.9	27.2	-3.5	27.8	-4.1	28.2	-4.9	28.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	10.4	12.7	17.9	23.7	29.0	30.7	29.6	28.6	27.3	23.4	17.4	12.3
	DBStd	7.66	2.43	3.53	4.05	3.59	3.08	2.55	2.06	1.74	1.95	2.56	2.55	2.64
	HDD10.0	43	25	11	1	0	0	0	0	0	0	0	0	7
	HDD18.3	702	246	161	58	3	0	0	0	0	1	48	186	
	CDD10.0	4407	37	85	245	412	589	620	607	576	518	417	221	79
	CDD18.3	2026	0	2	44	166	331	370	349	318	268	159	19	1
	CDH23.3	24681	1	27	419	1752	4481	5166	4538	3679	2933	1494	176	14
CDH26.7	12743	0	3	123	782	2577	3076	2429	1744	1415	576	17	1	
Wind	WSAvg	0.5	0.4	0.6	0.5	0.6	0.7	0.6	0.5	0.5	0.4	0.4	0.5	0.4
Precipitation	PrecAvg	417	12	22	30	36	27	51	105	76	36	12	4	7
	PrecMax	761	51	81	106	130	145	165	222	160	132	140	26	50
	PrecMin	101	0	0	0	0	0	3	15	5	0	0	0	0
	PrecStd	147	14	23	23	32	31	40	51	42	32	27	7	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	26.3	32.1	36.7	40.0	41.2	38.8	36.9	37.2	34.5	28.6	25.2
		MCWB	9.3	12.1	15.4	17.5	19.1	20.8	22.9	23.7	20.0	18.7	13.8	11.9
	2%	DB	19.8	23.8	29.9	34.6	38.5	39.7	37.4	35.6	35.3	33.0	26.7	22.6
		MCWB	8.9	11.6	15.1	17.3	18.1	20.6	23.0	23.2	20.3	18.5	13.9	10.6
	5%	DB	18.2	21.9	27.7	33.0	37.2	38.3	36.0	34.5	34.1	31.6	25.4	20.4
		MCWB	7.9	11.3	14.3	16.6	17.9	20.5	23.2	23.1	20.4	17.5	13.2	9.4
	10%	DB	16.6	19.7	25.5	31.1	35.8	36.8	34.7	33.4	32.9	29.9	24.0	18.7
		MCWB	7.5	10.1	13.1	16.2	17.5	20.2	23.1	23.2	20.6	16.5	12.6	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	14.6	17.5	20.9	21.9	25.6	25.8	26.2	25.1	21.6	16.9	13.2
		MCDB	19.8	22.1	28.8	31.7	35.3	35.9	34.2	34.0	31.4	29.6	23.3	23.5
	2%	WB	9.8	13.5	16.1	19.1	20.7	24.4	25.1	25.3	23.8	20.4	15.8	11.5
		MCDB	17.2	21.0	27.4	30.7	34.6	34.6	33.2	32.7	30.3	29.8	23.9	20.1
	5%	WB	8.9	12.3	14.9	17.9	19.9	23.5	24.6	24.6	22.9	19.4	14.7	10.3
		MCDB	16.0	20.3	24.9	29.9	33.1	33.6	32.7	31.9	29.5	28.8	23.3	18.6
	10%	WB	8.1	10.9	13.9	17.0	19.0	22.8	24.1	24.1	22.1	18.3	13.5	9.2
		MCDB	15.2	18.7	24.1	28.7	32.5	32.7	32.3	31.1	29.4	27.4	22.3	17.5
Mean Daily Temperature Range	5% DB	MDBR	11.6	11.3	11.4	11.5	12.0	11.7	10.4	9.9	11.3	12.4	12.0	11.9
		MCDBR	13.6	13.4	13.4	13.3	13.0	12.9	11.5	11.1	12.5	13.3	13.5	14.1
		MCWBR	6.4	6.1	5.2	5.1	4.5	4.5	3.7	3.7	4.4	5.1	6.2	7.1
	5% WB	MCDBR	12.2	11.9	12.2	12.3	12.7	12.2	10.8	10.3	10.8	12.6	12.3	12.9
		MCWBR	6.6	6.0	5.9	5.8	5.3	5.0	4.3	4.3	4.6	5.2	6.3	7.0
Clear-Sky Solar Irradiance	taub	0.375	0.420	0.447	0.502	0.524	0.633	0.719	0.639	0.506	0.454	0.432	0.375	
	taud	2.111	2.011	1.956	1.855	1.816	1.632	1.520	1.663	1.893	1.926	1.946	2.112	
	Ebn at Noon	836	830	836	803	786	700	640	688	772	782	760	813	
	Edh at Noon	132	159	180	206	216	258	287	247	189	171	154	126	
All-Sky Solar Radiation	RadAvg	3.65	4.39	5.67	6.65	7.25	6.96	6.33	6.09	5.89	5.16	4.02	3.50	
	RadStd	0.21	0.39	0.33	0.28	0.21	0.24	0.29	0.37	0.22	0.18	0.21	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page